

Another way to simplify our decision tree model is to do what's called *pruning*. For this, we grow a decision tree to its full depth and then find branches that we can remove from the tree without increasing our error rate by much.

Another refinement worth mentioning is using different measures to choose the right split point and the right splitting variable. In this chapter, we introduced the idea of using the classification error sum to decide where to put the split point; the right split point is one that minimizes our error sum. But there are other ways to decide on the right split point for a decision tree, including Gini impurity, entropy, information gain, and variance reduction. In practice, these other measures, especially Gini impurity and information gain, are almost always used rather than classification error rate, because some mathematical properties make them better in many cases. Experiment with different ways to choose a split point and splitting variable to find one that seems to perform the best for your data and your decision problem.

Everything we do in machine learning is meant to enable us to make accurate predictions on new data. When you're trying to improve a machine learning model, you can always judge whether an action is worthwhile by checking how much it improves your error rate on test data. And feel free to be creative to find improvements—anything that improves your error rate on test data is probably worth trying.

Random Forests

Decision trees are useful and valuable, but they are not regarded as the best machine learning method by professionals. This is in part because of their reputation for overfitting and relatively high error rates, and in part because of the invention of a method called *random forests*, which has become popular recently and provides an unequivocal performance improvement over decision trees.

As its name suggests, a random forest model consists of a collection of decision tree models. Each decision tree in the random forest depends on some randomization. Using randomization, we get a diverse forest with many trees instead of a forest that is just one tree repeated over and over.